

1 · What Makes an AI Image Detectable?

Real cameras and generative models leave different fingerprints — detectors learn to spot them:

- **Frequency artifacts:** GAN/diffusion up-sampling leaves periodic patterns in the Fourier spectrum that cameras don't produce.
- **Noise & sensor fingerprints:** real photos carry sensor noise (PRNU); synthetic images lack it or fake it imperfectly.
- **Texture & fine detail:** skin, hair, foliage, text and reflections are where models still slip.
- **Semantic / physics tells:** impossible lighting, warped hands, garbled text, inconsistent shadows.

Key Insight: Go hybrid

- Best detectors combine **high-level CLIP semantics** + **low-level frequency patches** — each catches what the other misses, and both survive different transforms.
- Many signals live in **high-frequency detail** — exactly what compression and blur destroy. That's why robustness is hard.
- **Don't just fine-tune a classifier.** Think about what your model is actually learning — is it a real artifact, or a dataset shortcut?

2 · A Baseline Detection Pipeline

Start simple. A clean binary classifier you can actually finish beats a fancy model you can't.



- Fine-tune a pretrained backbone (ResNet / EfficientNet / ViT) as a binary classifier — a strong day-1 baseline.
- **Optional upgrade:** add a frequency branch (FFT / DCT features) and fuse it with the spatial branch.
- Output a **calibrated probability**, not just a label — you'll need it for thresholding & error analysis.

3 · The Key Idea: Train for the Real World

The biggest lever isn't a fancier model — it's **what you train on**.



- **Augmentation = simulate redistribution during training:** JPEG-compress, blur, resize, crop, colour-jitter, add noise, re-screenshot.
- **SAFE insight (KDD 2025):** crop instead of down-sample to preserve high-freq artifacts; ColorJitter + RandomRotation kill colour/semantic shortcuts.
- **DDA insight (NeurIPS 2025):** watch out for frequency bias — JPEG in your real images can become a spurious signal. Align pixel + frequency.
- **Augmentation + data alignment > architecture tricks.** Training-pipeline improvements beat fancier backbones.

Mental model: "If a transformation can happen on a real feed, it must happen in your training pipeline."

Augmentation Parameters

Apply these **randomly during training** — they simulate the real-world redistribution pipeline. Required transforms are tested at evaluation.

Transform	Parameters	Real-World Analog
JPEG Compression	quality = 90, 70, 50, 30	Social-media re-encode, messaging
Gaussian Blur	kernel σ = 0.5, 1.0, 2.0	Out-of-focus, screenshot smoothing
Resize	scale $0.5\times$ / $0.25\times$ then upscale	Thumbnail generation, CDN resize
Gaussian Noise	σ = 0.02, 0.05, 0.10	Low-light sensor noise
Color Jitter	brightness/contrast/sat. $\pm 20\%$	Filter apps, auto-enhance
Center Crop	crop 80%	Profile-picture cropping, framing

4 · Evaluate Like It's the Real World

Clean accuracy is the #1 way to fool yourself.

- Build a **transformed test set**: e.g., JPEG q=90/70/50/30.
- **Primary metric**: ROC AUC (threshold-free, robust to imbalance).
- **Final Score** = $0.50 \times \text{AUC}_{\text{clean}} + 0.50 \times \text{AUC}_{\text{robust}}$
- **Cross-generator**: test on generators NOT in training — the real generalization test.

Condition	Acc.	AUC
Clean	0.97	0.99
JPEG q30	0.86	0.93
Blur $\sigma=2$	0.82	0.90
Crop 80%	0.88	0.94
Unseen gen.	0.71	0.80

Judges want a compact robustness table + error-analysis note.